

Original Research Article

MGIT sensitivity testing and genotyping of drug resistant *Mycobacterium tuberculosis* isolates from Mizoram, Northeast India

Christine Vanlalbiakdiki Sailo^{a,b}, Zothan Zami^a, Ralte Lalremruata^b, Zothan Sanga^c,
Vanlal Fela^c, Febiola Kharkongor^c, Lily Chhakchhuak^d, Zothankhuma Chhakchhuak^e,
Gracy Laldinmawii^f, Dhiraj Kumar^g, Nachimuthu Senthil Kumar^{a,*}

^a Department of Biotechnology, Mizoram University, Aizawl 796004, Mizoram, India

^b Department of Microbiology, Synod Hospital, Durtlang, Aizawl 796025, Mizoram, India

^c Department of Health and Family Welfare, Directorate of Health Services, Aizawl 796009, Mizoram, India

^d Directorate of Health Services, National Health Mission, Aizawl 796009, Mizoram, India

^e CMO, Health & Family Welfare Department, Mamit 796441, Mizoram, India

^f Department of Microbiology, Zoram Medical College, Falkawn 796005, Mizoram, India

^g Cellular Immunology Group, International Centre for Genetic Engineering & Biotechnology, Aruna Asaf Ali Marg, New Delhi 110067, India

ARTICLE INFO

Keywords:
Mizoram
MGIT 960
GeneXpert
Sanger sequencing
Mtb lineages

ABSTRACT

Purpose: Tuberculosis, a crucial infectious disease is still a health concern globally. India is among the countries with high MDR-TB burden. Currently, sputum smear microscopy using Ziehl Neelsen stain and GeneXpert are the only diagnostic means in Mizoram. This study was done to characterize local tuberculosis strains circulating in Mizoram.

Methods: Sputum was cultured using MGIT 960 and DST was performed for Streptomycin, Isoniazid, Rifampicin, Ethambutol and Pyrazinamide. GeneXpert test was done simultaneously. DNA was extracted using Trueprep AUTO v2, molbio diagnostics. Antibiotic Resistance Genes and LSP were amplified and sequenced.

Results: Ser315Thr was the most common mutation in *katG* among MDR-TB isolates. GeneXpert probes A and D drop out upon sequencing showed L511P, H526Q and H526L mutation. The L511P and H526Q mutations were seen in new and treated cases. Discrepancy between MGIT 960 and GeneXpert were observed. LSP-PCR revealed that Indo-Oceanic, East-African Indian, Euro-American and Beijing lineages were found in Mizoram.

Conclusion: This study provides mutation information on the resistant genotypes detected with GeneXpert as well as MGIT 960. It also provides information on the lineages of *Mycobacterium tuberculosis* circulating in the state. Utilization of sequencing technologies is essential in diagnostic laboratories to rule out discrepant results and as a cautionary measure to prevent wrong diagnosis and treatment.

1. Introduction

Tuberculosis which is still a global health concern is caused by *Mycobacterium tuberculosis complex* (MTBc) comprising of human and animal adapted pathogens [1]. Resistance to different anti-tuberculosis drugs has been attributed to mutations in specific genes [2]. With the advent of GeneXpert, primary screening involves PCR-based detection of MTB DNA as well as a targeted SNP based PCR in the *rpoB* (RIF resistance) gene. The rationale behind using *rpoB* SNP is that it serves as a surrogate marker for Rifampicin, since >90% of Rifampicin resistant isolates also exhibit Isoniazid resistance [3]. Thus, a positive result

indicates confirmation of tuberculosis as well as multi-drug resistant infection.

MTB phylogeny consists of six major lineages (L) named according to their geographic distribution: L1 to L6 [4]. Few other lineages were recently identified in the African continent [5]. Previous studies have shown varied disease progression and relapse risk in different lineages [6]. MTB strains harbour insertions or deletions in the genome called Large Sequence Polymorphism (LSPs), also known as Regions of Difference (RDs) which are unique polymorphisms with mutations occurring only once in a species. Thus, RDs are irreversible and do not show homoplasmy, hence have been used to differentiate MTB into its respective lineages [7].

* Corresponding author.

E-mail address: nskbiotech@mzu.edu.in (N.S. Kumar).

<https://doi.org/10.1016/j.ijmm.2022.06.003>

Received 19 January 2022; Received in revised form 17 May 2022; Accepted 13 June 2022

Available online 25 June 2022

0255-0857/© 2022 Indian Association of Medical Microbiologists. Published by Elsevier B.V. All rights reserved.

India is among the 3 countries with high Multidrug Resistant (MDR) TB burden and contributes largely to the global increase in TB [8]. Mizoram having a population of 12.5 lakhs, the total TB patients notified from both public and private sector in 2019 was 2944 [9]. Currently, sputum smear microscopy using Ziehl Neelsen stain and Cartridge Based Nucleic Acid Amplification Test (CB-NAAT) or GeneXpert are the only diagnostic means implemented in the state of Mizoram.

This study aims to characterize the MTB strains which will help to establish: 1) whether initial infection of individuals could result from a heterogeneous mixture of MTB strains; 2) the status of MDR and the associated variants; and 3) the lineage of the MTB strains present in the region. This is the first study from Northeast India on the extent of heterogeneity in the initial acquired infections which might prove great importance in the designing of informed treatment regimen for tuberculosis patients as well as future monitoring of the MDR/virulent strains in the region.

2. Materials and methods

2.1. Sample collection

Participants were recruited from District TB Center, Falkawn and Synod Hospital, Durtlang, Mizoram during 2019–2020. This includes new and previously treated pulmonary TB cases, aged 15 years and above, both genders without HIV infection. A total of 88 sputum smear positive samples were collected in a screw capped container.

2.2. GeneXpert test

Xpert MTB/RIF assay G4 (Cepheid) was performed in accordance to the manufacturer's instructions with a reported sensitivity of 94.4% and specificity of 98.3% for diagnosis of rifampicin resistance [3]. Presence of *rpoB* mutations can change the dynamics of hybridization, resulting in

differences between the Ct values of the probes. Inhibition of hybridization results in probe dropout. However, in samples without mutated *rpoB*, all the 5 probes match exactly to the amplified MTB DNA. The G4 version of Xpert MTB/RIF software interprets sample results as resistant to Rifampicin, if the difference between two probes (the first and last Ct) is >4 cycles ($\Delta Ct > 4$) [10].

2.3. Culture and drug susceptibility test using MGIT 960

Culture of MTB from sputum was performed using automated liquid culture BACTEC MGIT 960 system (BD, USA) in accordance with FIND Diagnostics. The workflow for selection of samples is shown in Fig. 1. Drug susceptibility testing (DST) was performed from growth positive tubes using first line drugs - Streptomycin (1.00 µg/ml), Isoniazid (0.10 µg/ml), Rifampicin (1.00 µg/ml) and Ethambutol (5.00 µg/ml) using BD BACTEC MGIT™ 960 SIRE Kit and Pyrazinamide (100 µg/ml), provided the cultures are young i.e., one to two days old. From the third day onwards to the fifth day, a 1:5 dilution was employed [11] (see Fig. 2). Subculture was also maintained on LJ media using Lowenstein-Jensen Medium Slants for DNA extraction and further molecular testing (Supplementary Table 1).

2.4. DNA extraction

DNA was extracted from 44 LJ cultures. A loop full of the growth was emulsified in 2 ml of 0.9% normal saline, vortexed and heat killed at 80 °C for 20 min [12]. It was centrifuged in a cooling centrifuge at 14,000 rpm for 5 min, the supernatant discarded and the pellet was emulsified in 1 ml of 7H9 Broth. Lysis buffer was added and finally transferred into Universal Cartridge (Trueprep) and loaded into nucleic acid extractor machine (Trueprep AUTO v2). Extraction was done as per manufacturer's instructions.

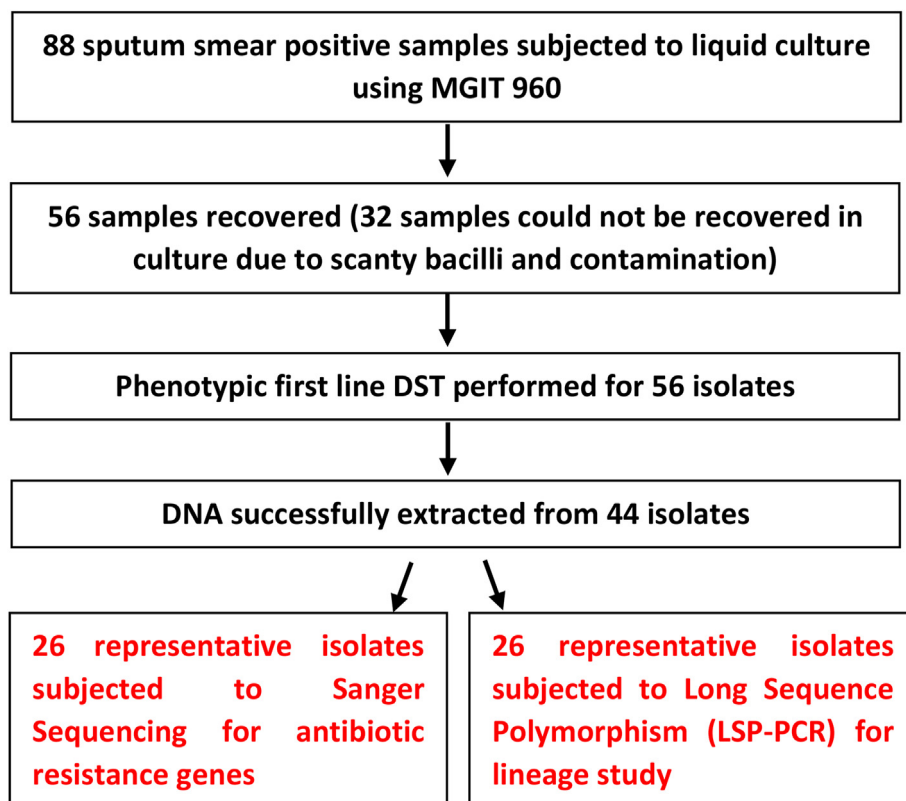


Fig. 1. Flowchart representing the workflow for selection of samples

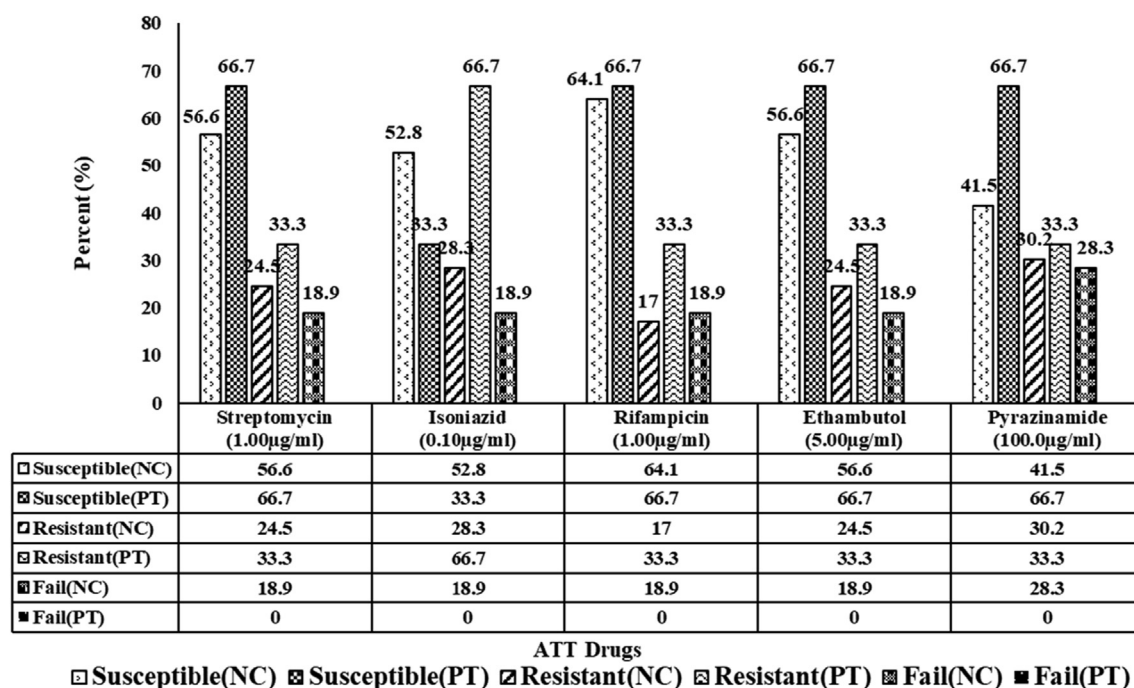


Fig. 2. Drug Susceptibility of new and previously treated cases for first line anti-tuberculosis drugs using MGIT 960. NC-New Case; PT- Previously Treated.

2.5. DNA quantitation, PCR amplification and sequencing

The isolated DNA was checked on a 0.8% agarose gel electrophoresis (BioRad), quantified using a µdrop MultiScan SkyHigh Spectrophotometer (ThermoScientific). Polymerase Chain Reaction (PCR) was run on five antibiotic resistance genes towards first line ATT (Anti-tuberculosis treatment) drugs. The primer sequences and conditions for each primer is given in [Supplementary Table 2](#). PCR was carried out in a 10 µl reaction and the PCR product was observed in a 1.5% Agarose Gel Electrophoresis (Bio Rad). Sequencing was done using 3500 Genetic Analyzer automated sequencer (ABI), analyzed for confirmation and presence of variants in the genes. The sequences were aligned using Clustal Omega with reference sequences (*Mycobacterium tuberculosis* H37Rv) obtained from Mycobrowser [13].

2.6. Genotyping of *Mycobacterium tuberculosis* isolates from Mizoram

Genotyping of MTB lineage was done using Long Sequence Polymorphism (LSP-PCR) otherwise known as Regions of Differences (RDs). Based on LSP (RD), the major MTB lineages were classified using four primers - Indo-Oceanic (L1), Beijing (L2), East-African Indian (L3) and Euro-American (L4). The primers and PCR conditions used for classification of lineages are shown in [Supplementary Table 2](#).

3. Results

3.1. Drug susceptibility report

The phenotypic drug susceptibility for 56 samples ([Fig. 1](#)) was interpreted into three measures as: Resistant, Sensitive and DST failed. Isoniazid (30.4%) and pyrazinamide (30.4%) showed the highest drug resistance, Streptomycin and ethambutol showed similar resistance pattern (25.0%), while rifampicin showed the lowest resistance rate (17.9%). In addition, phenotypic drug susceptibility test also failed by a proportion of 17.9% for streptomycin, isoniazid, rifampicin and ethambutol while pyrazinamide failed by 26.8%.

3.2. Isoniazid (INH)

Out of the 26 representative samples, Phenotypic DST (MGIT) showed 34.6% resistance to INH and sequencing showed that 70% out of the resistant samples harboured SNP at codon 315 *katG* gene. Among these, the most common amino acid change observed was the substitution of Serine to Threonine (85.7%) occurring at (AGC > ACC). A single sample (14.3%) had a Ser315Asn substitution (AGC > AAC). Mutations were not found in any of the INH susceptible isolates and also in three INH resistant samples ([Table 1](#)).

3.3. Rifampicin (RIF)

The GeneXpert results showed that out of the 26 samples, two samples had probes dropout at both probes A (codons 507–511) and D (codons 523–529). A single probe A dropout was observed in 11.5% of samples and 3.8% of samples had a single probe drop out at D ([Table 1](#)).

Sequencing revealed an amino acid substitution from Leucine to Proline (L511P, CTG → CCG) at probe A (codons 507–511) in the samples with Probe A drop out. The most common mutation occurring at probe D (codons 523–529) was an amino-acid substitution from Histidine to Glutamine (H526Q, CAC → CAG). A less common mutation observed for one sample at probe D was amino-acid substitution from Histidine to Leucine (H526L, CAC → CTC). For one sample (CVS-3[#]), both isoniazid (via. MGIT 960) and rifampicin (via Xpert MTB/RIF) were resistant, hence an MDR. However, in GeneXpert there was no probe drop out in all the five probes. The ΔCt was >4 resulting in the instrument calling it a rifampicin resistance. In addition, for this particular sample, the phenotypic DST for Rifampicin performed using MGIT had shown it as susceptible ([Table 1](#)).

Apart from the concordant phenotypic (MGIT 960) and genotypic (Xpert MTB/RIF) results, discordant DST results for Rifampicin were also observed for 7 samples: 6 samples showed as MGIT 960 susceptible while the Xpert MTB/RIF shows resistant. 3 samples were resistant to rifampicin via. MGIT 960 and susceptible via Xpert MTB/RIF. These results are shown in [Supplementary Tables 3 and 4](#).

Table 1
Antibiotic resistance and genes associated with variants and Lineage of MTB.

Sample Code	Isoniazid			Rifampicin							LSP markers		
				Drug Susceptibility Test		GeneXpert Probes Result							Sanger Sequencing
	Pheno typic DST (MGIT)	katG codon change (G/C/A)	Amino acid change	Pheno typic DST (MGIT)	Genotypic Rif-Resistance (GeneXpert)	A (507–511)	B (512–518)	C (518–523)	D (523–529)	E (529-533)	SNP	Amino acid change	Lineage
CVS-1	R	AGC	Ser315	R	Not detected	+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/EuA
CVS-3 [#]	R	AAC	Ser315Asn	S	Detected	+	+	+	+	+	No SNP detected	No SNP detected	Non-Beijing/non-IO/non-EAI/EuA
CVS-6 [#]	R	ACC	Ser315Thr	R	Detected	Neg	+	+	Neg	+	CTG > CCG, CAC > CAG	L511P, H526Q	Non-Beijing/non-IO/non-EAI/non-EuA
CVS-9 [#]	S	AGC	Ser315	S	Detected	+	+	+	+	+	No SNP detected	No SNP detected	Non-Beijing/non-IO/non-EAI/non-EuA
CVS-13	R			R	Not detected	+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
CVS-21	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
CVS-24	S			S		+	+	+	+	+	–	–	Non-Beijing/IO/EAI/non-EuA
CVS-25	S			S		+	+	+	+	+	–	–	Non-Beijing/IO/non-EAI/non-EuA
NER-4	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
NER-10	R	ACC	Ser315Thr	R	Detected	Neg	+	+	+	+	CTG > CCG	L511P	Non-Beijing/non-IO/non-EAI/non-EuA
NER-13	S	AGC	Ser315	S	Not detected	+	+	+	+	+	–	–	Non-Beijing/IO/non-EAI/non-EuA
NER-19	S			S		+	+	+	+	+	–	–	Non-Beijing/IO/non-IO/non-EAI/EAI/non-EuA
NER-23	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
NER-24	R	ACC	Ser315Thr	S	Detected	Neg	+	+	+	+	CTG > CCG	L511P	Non-Beijing/non-IO/non-EAI/non-EuA
NER-32	S	AGC	Ser315	S	Not detected	+	+	+	+	+	–	–	Beijing/non-IO/non-EAI/non-EuA
NER-35	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/EAI/non-EuA
NER-36	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
NER-37	R	ACC	Ser315Thr	R	Detected	+	+	+	Neg	+	CAC > CTC	H526L	Non-Beijing/non-IO/non-EAI
NER-41	S	AGC	Ser315	S	Not detected	+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA

(continued on next page)

Table 1 (continued)

Sample Code	Isoniazid			Rifampicin								LSP markers	
				Drug Susceptibility Test		GeneXpert Probes Result					Sanger Sequencing		Lineage
	Pheno typic DST (MGIT)	katG codon change (G/C/A)	Amino acid change	Pheno typic DST (MGIT)	Genotypic Rif-Resistance (GeneXpert)	A (507–511)	B (512–518)	C (518–523)	D (523–529)	E (529–533)	SNP	Amino acid change	
NER-42	R	ACC	Ser315Thr	S	Detected	Neg	+	+	+	+	CTG > CCG	L511P	Non-Beijing/non-IO/non-EAI/non-EuA
NER-43	S	AGC	Ser315	S	Not detected	+	+	+	+	+	–	–	Non-Beijing/IO/non-IO/non-EAI/non-EuA
NER-44	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
NER-45	S			S		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/EuA
NER-46	R	ACC	Ser315Thr	R	Detected	Neg	+	+	Neg	+	CTG > CCG, CAC > CAG	L511P, H526Q	Non-Beijing/non-IO/non EAI/EAI/non-EuA
CVS-4	NA	AGC	Ser315	NA	Not detected	+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA
CVS-17	R			R		+	+	+	+	+	–	–	Non-Beijing/non-IO/non-EAI/non-EuA

S- Susceptible, R- Resistant; NA- Not available; + positive for GeneXpert probe; Neg-GeneXpert probe dropout; # previously treated cases; **Non-Beijing** (435 bp); **Beijing** (absence of 435 bp); **IO** Indo-Oceanic (210 bp); **non-IO** non Indo-Oceanic (1053 bp); **EAI** East Africa Indian (158 bp); **non-EAI** non-East African Indian (948 bp); **EuA** Euro-American (absence of 332 bp); **non-EuA** non Euro-American (332 bp).

3.4. Streptomycin, Ethambutol and Pyrazinamide

Phenotypic DST reported resistance for Streptomycin (14 samples), Ethambutol (14 samples) and Pyrazinamide (17 samples) (Supplementary Table 1). There were no mutations found in the genes corresponding to these antibiotics.

3.5. Lineage of *M. tuberculosis*

About 11.5% of the samples were Indo-Oceanic (L1), 7.7% samples were East-African Indian (L3) and 11.5% samples were Euro-American (L4). We also identified mixed infection of both Indo-Oceanic and East African Indian in two samples (7.7%). About 57.7% of samples which were not identified among the mentioned lineages were reported as non-Beijing, non- Indo-Oceanic, non-East-African India and non-Euro-American. A single sample out of the 26 samples belonged to Beijing (L2) (Table 1).

4. Discussion

This is the first attempt to culture tubercle bacilli in Mizoram, since there are no functional MTB culture facilities implemented. Limited scientific publications on MTB and drug resistance from this region necessitates this study. INH requires activation by *katG* protein to exert its toxic effect on the bacillus. More than 300 *katG* mutations have been identified, however codon 315 (serine to threonine) accounts for 95% mutation [14]. In this study, similar report was observed where the 7-INH resistant isolates which are also Rifampicin resistant had the most common substitution as Ser315Thr ($n = 6/7$, 85.7%) in *katG* gene. It is also admissible that the Ser315Thr substitution is associated with MDR strains [15]. A less common Ser315Asn substitution found in this study was also reported by a study from India [16]. The three samples that had a phenotypic INH resistance but is not reflected in the sanger report might possibly be due to the INH resistance mutation occurring in other genes such as *inhA* [14].

Rifampicin discordant results observed between MGIT 960 and Xpert MTB/RIF in 3 isolates (Supplementary Table 3) in which mutations were absent in the RRDR region of the *rpoB* gene though phenotypically resistant to RIF. This might be attributed to different genotypes prevailing worldwide and since Xpert MTB/RIF detects mutations only within the RRDR of the *rpoB* gene, the mutations in these strains might lie outside this region as described previously [17]. In two samples (CVS-3[#], CVS-9[#]), we observed a discrepancy between MGIT and Xpert MTB/RIF where rifampicin was susceptible via. MGIT, resistant via. Xpert MTB/RIF without any probe drop out and ΔC_t was >4 , however sanger sequencing revealed no mutation in *rpoB* region. This could be a false resistance produced by Xpert MTB/RIF as similar discrepancy results were reported from China [18]. In two other samples (NER-24, NER-42), rifampicin was susceptible via. MGIT and resistant via. Xpert MTB/RIF where probe A was drop out. In this discrepancy, the mutation observed was L511P (CTG $\rightarrow$ CCG). Similar disputed *rpoB* mutation was reported from South Korea [19]. A study from Kuwait reported isolates as rifampin susceptible in MGIT, but resistant by Xpert assay. Sequencing of *rpoB* identified a silent (CTG521TTG) and a missense (GAC516TAC) mutation [20]. However, this was not the case in our study, suggesting diverse as yet unknown mechanisms, impacting rifampin susceptibility.

In this region, sanger sequencing of the specific mutation in probe A (507–511) was L511P. Our study also reported the specific mutation occurring at probe D (523–529) as H526Q and H526L which is not in concordance with another study from Bangladesh where they reported mutation as H526D and H526Y [21]. The mutations L511P and H526L have been reported to confer a low level of resistance [22]. A study from Treichville, Abidjan District reported H526Q mutation in a relapse case [23]. With respect to the other antibiotics tested, the phenotypic DST resistance observed for Streptomycin, Ethambutol and Pyrazinamide do not correlate with sanger sequencing results, probably because the

primers used in this study does not cover the region where SNP's might be present.

Though the number of samples studied maybe small, this study attempts to provide information on *M. tuberculosis* strain diversity in this remote region of northeast India where all the four major lineages (L1, L2, L3 L4) are present. In a recent study which is under review, Beijing (L2) and East-African Indian (L3) were found in three of the northeastern states, Manipur, Nagaland and Tripura [24]. Studies have revealed Indo-Oceanic (L1) and East-African Indian (L3) to be most widespread in India compared to other parts of the world [25]. This might explain the mixed strain infection observed in two samples in this study.

Mixed strain infections particularly when strains with different drug resistance phenotypes are involved, makes diagnosis, treatment and control more cumbersome and people with underlying morbidity may be prone to mixed strain infection [26]. A study from Botswana had reported that among new cases of tuberculosis, mixed strain infections (but not hetero-resistant infection) were associated with poor treatment outcomes using standard combination treatment regimens [27]. Previous studies have shown that in patients who harboured both MDR and susceptible strains, MDR strain was able to persist and grow during treatment with first-line regimens and upon a switch to second-line regimen, the susceptible strain was able to re-emerge, and thereby compromising treatment outcome [28]. Immune responses may be altered in patients with ongoing mixed strain TB infections, rendering them more prone to re-infection [29].

In Mizoram, establishment of highly sophisticated diagnostic machines requiring technical expertise is still cumbersome. Due to programmatic challenges, ATT is usually initiated based on GeneXpert reports. Since the global frontline molecular diagnostic tools including GeneXpert and LPA are based on inadequate number of known genetic markers, seldom, phenotypic resistance could not be found associated with known drug resistant mutations [30]. Thus, our findings suggest that routine diagnosis aided with sequence-based information is essential in diagnostic laboratories to map the genetic variations and identify all the genotypes associated with drug resistance to provide newer, accurate diagnosis, better anti-tubercular therapies, vaccines and effective patient care.

5. Conclusion

This study provides information on the specific mutations detected with GeneXpert and MGIT 960 as well as the lineages of MTB circulating in the state. Being the first MTB culture to be performed within the state, the phenotypic DST is yet to be certified since it is not from a competent laboratory. However, from our small set of samples, we report fundamental data that represents the present status of the disease in the region that can be further explored by expanding the genetic regions covered for the antibiotic resistance genes as well as stronger lineage identification markers with larger number of samples.

Authors Contribution

VF, FK, ZC, LC recruited the patients. CVS, RL, ZS, GL collected the samples and performed the culture. CVS, ZS, ZC, LC, FK, ZZ performed the MGIT and Xpert assays, molecular work and data analysis. CVS, ZZ, VF, NSK interpreted the data and wrote the manuscript. NSK, RL, GL, ZS, DK contributed to the concept and design of the study. NSK, DK, VF acquired the funds for the study. All authors have proof read and approved the manuscript.

Funding

This study was funded by Department of Biotechnology (DBT), New Delhi, Govt. of India under MDR -TB NER Project (BT/PR23092/NER/95/589/2017).

Declaration of competing interest

The authors declare no conflict of interest.

Ethics approval

Approval for the study was obtained from the Institutional Ethics Committee (IEC) of Civil Hospital, Aizawl, Mizoram (B.12,018/1/13-CH (A)/IEC/63). All the participants had provided a written consent at the time of sample collection.

Acknowledgements

The authors would like to thank all the participants of this study and the laboratory technologists who assisted in sample collection - Ms. Jennifer Lalrammawii, Mr. Lalramnunsiana and Mr. Lalhmangaihsanga. We also thank the District TB Center staff of Mizoram for their kind cooperation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmm.2022.06.003>.

References

- Pesciaroli M, Alvarez J, Boniotti MB, Cagiola M, Di Marco V, Marianelli C, et al. Tuberculosis in domestic animal species. *Res Vet Sci* 2014;97(Suppl):S78–85. <https://doi.org/10.1016/j.rvsc.2014.05.015>.
- Tekwu EM, Sidze LK, Assam JP, Tedom JC, Tchatchouang S, Makafe GG, et al. Sequence analysis for detection of drug resistance in *Mycobacterium tuberculosis* complex isolates from the Central Region of Cameroon. *BMC Microbiol* 2014;3(14): 113. <https://doi.org/10.1186/1471-2180-14-113>.
- Boehme CC, Nicol MP, Nabeta P, Michael JS, Gotuzzo E, Tahirli R, et al. Feasibility, diagnostic accuracy, and effectiveness of decentralised use of the Xpert MTB/RIF test for diagnosis of tuberculosis and multidrug resistance: a multicentre implementation study. *Lancet* 2011;377(9776):1495–505. [https://doi.org/10.1016/S0140-6736\(11\)60438-8](https://doi.org/10.1016/S0140-6736(11)60438-8).
- Nicol MP, Sola C, February B, Rastogi N, Steyn L, Wilkinson RJ. Distribution of strain families of *Mycobacterium tuberculosis* causing pulmonary and extrapulmonary disease in hospitalized children in Cape Town, South Africa. *J Clin Microbiol* 2005; 43(11):5779–81. <https://doi.org/10.1128/JCM.43.11.5779-5781.2005>.
- Coscolla M, Gagneux S, Menardo F, Loiseau C, Ruiz-Rodriguez P, Borrell S, et al. Phylogenomics of *Mycobacterium africanum* reveals a new lineage and a complex evolutionary history. *Microb Genom* 2021;7(2):000477. <https://doi.org/10.1099/mgen.0.000477>.
- Burman WJ, Bliven EE, Cowan L, Bozeman L, Nahid P, Diem L, et al. Tuberculosis Trials Consortium. Relapse associated with active disease caused by Beijing strain of *Mycobacterium tuberculosis*. *Emerg Infect Dis* 2009;15(7):1061–7. <https://doi.org/10.3201/eid1507.081253>.
- Hirsh AE, Tsolaki AG, DeRiemer K, Feldman MW, Small PM. Stable association between strains of *Mycobacterium tuberculosis* and their human host populations. *Proc Natl Acad Sci U S A* 2004;101(14):4871–6. <https://doi.org/10.1073/pnas.0305627101>.
- Global tuberculosis report 2020. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.
- India TB Report 2020. <https://tbcindia.gov.in> WriteReadData India TB. Accessed on 15 September 2021.
- Ocheretina O, Byrt E, Mabou MM, Royal-Mardi G, Merveille YM, Rouzier V, et al. False-positive rifampin resistant results with Xpert MTB/RIF version 4 assay in clinical samples with a low bacterial load. *Diagn Microbiol Infect Dis* 2016;85(1): 53–5. <https://doi.org/10.1016/j.diagmicrobio.2016.01.009>.
- Siddiqi SH, Rusch-Gerdes S. MGIT™ procedure manual-FindDx.org. 2006. http://www.finddx.org/wp-content/uploads/2016/02/mgit_manual_nov2006.pdf.
- Doig C, Seagar AL, Watt B, Forbes KJ. The efficacy of the heat killing of *Mycobacterium tuberculosis*. *J Clin Pathol* 2002;55(10):778–9. <https://doi.org/10.1136/jcp.55.10.778>.
- Sievers F, Wilm A, Dineen D, Gibson TJ, Karplus K, Li W, et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol* 2011;7:539. <https://doi.org/10.1038/msb.2011.75>.
- Seifert M, Catanzaro D, Catanzaro A, Rodwell TC. Genetic mutations associated with isoniazid resistance in *Mycobacterium tuberculosis*: a systematic review. *PLoS One* 2015;10(3):e0119628. <https://doi.org/10.1371/journal.pone.0119628>.
- Marttila HJ, Soini H, Eerola E, Vyshnevskaya E, Vyshnevskiy BI, Otten TF, et al. A Ser315Thr substitution in KatG is predominant in genetically heterogeneous multidrug-resistant *Mycobacterium tuberculosis* isolates originating from the St. Petersburg area in Russia. *Antimicrob Agents Chemother* 1998;42(9):2443–5. <https://doi.org/10.1128/AAC.42.9.2443>.
- Jaiswal I, Jain A, Singh P, Verma SK, Prakash S, Dixit P, et al. Mutations in katG and inhA genes of isoniazid-resistant and -sensitive clinical isolates of *Mycobacterium tuberculosis* from cases of pulmonary tuberculosis and their association with minimum inhibitory concentration of isoniazid. *Clin Epidemiol Glob Health* 2017; 5(3):143–7. <https://doi.org/10.1016/j.cegh.2016.08.008>.
- Heep M, Brandstätter B, Rieger U, Lehn N, Richter E, Risch-Gerdes S, et al. Frequency of rpoB mutations inside and outside the cluster I region in rifampin-resistant clinical *Mycobacterium tuberculosis* isolates. *J Clin Microbiol* 2001;39(1): 107–10. <https://doi.org/10.1128/JCM.39.1.107-110.2001>.
- Qin L, Huo F, Ren W, Shang Y, Yao C, Zhang X, et al. Dependence of Xpert MTB/RIF accuracy for detecting rifampin resistance in bronchoalveolar lavage fluid on bacterial load: a retrospective study in Beijing, China. *Infect Drug Resist* 2021;14: 2429–35. <https://doi.org/10.2147/IDR.S307488>.
- Jo KW, Lee S, Kang MR, Sung H, Kim MN, Shim TS. Frequency and type of disputed rpoB mutations in *Mycobacterium tuberculosis* isolates from South Korea. *Tuberc Respir Dis* 2017;80(3):270–6. <https://doi.org/10.4046/trd.2017.80.3.270>.
- Mokaddas E, Ahmad S, Eldeen HS, Al-Mutairi N. Discordance between Xpert MTB/RIF assay and Bactec MGIT 960 Culture System for detection of rifampin-resistant *Mycobacterium tuberculosis* isolates in a country with a low tuberculosis (TB) incidence. *J Clin Microbiol* 2015;53(4):1351–4. <https://doi.org/10.1128/JCM.03412-14>.
- Uddin MKM, Rahman A, Ather MF, Ahmed T, Rahman SMM, Ahmed S, et al. Distribution and frequency of rpoB mutations detected by Xpert MTB/RIF assay among Beijing and non-Beijing rifampicin resistant *Mycobacterium tuberculosis* isolates in Bangladesh. *Infect Drug Resist* 2020;13:789–97. <https://doi.org/10.2147/IDR.S240408>.
- Hauck Y, Fabre M, Vergnaud G, Soler C, Pourcel C. Comparison of two commercial assays for the characterization of rpoB mutations in *Mycobacterium tuberculosis* and description of new mutations conferring weak resistance to rifampicin. *J Antimicrob Chemother* 2009;64(2):259–62. <https://doi.org/10.1093/jac/dkp204>.
- N'guessan Kouassi K, Riccardo A, Dutoziet Christian C, André G, Férlaha C, Hortense SA, et al. Genotyping of mutations detected with GeneXpert. *Int J Mycobacteriol* 2016;5(2):142–7. <https://doi.org/10.1016/j.ijmyco.2016.01.001>.
- Dusthacker A, Kumar A, Mohanvel SK, Mahizhavan B, Shivakumar S, Raghavi S, et al. *Mycobacterium tuberculosis* strain lineage in mixed tribal population across India and Andaman Nicobar Island. *World J Microbiol Biotechnol* 2021;37(11):192. <https://doi.org/10.1007/s11274-021-03164-6>.
- Ahmed N, Ehtesham NZ, Hasnain SE. Ancestral *Mycobacterium tuberculosis* genotypes in India: implications for TB control programmes. *Infect Genet Evol* 2009;9(1):142–6. <https://doi.org/10.1016/j.meegid.2008.10.001>.
- Asare-Baah M, Séraphin MN, Salmon LAT, Lauzardo M. Effect of mixed strain infections on clinical and epidemiological features of tuberculosis in Florida. *Infect Genet Evol* 2021;87:104659. <https://doi.org/10.1016/j.meegid.2020.104659>.
- Shin SS, Modongo C, Baik Y, Allender C, Lemmer D, Colman RE, et al. Mixed *Mycobacterium tuberculosis*-strain infections are associated with poor treatment outcomes among patients with newly diagnosed tuberculosis, independent of pretreatment heteroresistance. *J Infect Dis* 2018;218(12):1974–82. <https://doi.org/10.1093/infdis/jiy480>.
- van Rie A, Victor TC, Richardson M, Johnson R, van der Spuy GD, Murray EJ, Beyers N, Gey van Pittius NC, van Helden PD, Warren RM. Reinfection and mixed infection cause changing *Mycobacterium tuberculosis* drug-resistance patterns. *Am J Respir Crit Care Med* 2005;172(5):636–42. <https://doi.org/10.1164/rccm.200503-449OC>. Epub 2005 Jun 9. PMID: 15947286; PMCID: PMC2718533.
- Warren RM, Victor TC, Streicher EM, Richardson M, Beyers N, Gey van Pittius NC, van Helden PD. Patients with active tuberculosis often have different strains in the same sputum specimen. *Am J Respir Crit Care Med* 2004;169(5):610–4. <https://doi.org/10.1164/rccm.200305-714OC>. Epub 2003 Dec 30. PMID: 14701710.
- Schön T, Miotto P, Köser CU, Viveiros M, Böttger E, Cambau E. *Mycobacterium tuberculosis* drug-resistance testing: challenges, recent developments and perspectives. *Clin Microbiol Infect* 2017;23(3):154–60. <https://doi.org/10.1016/j.cmi.2016.10.022>.